Table S1: SMILES of generated and well-known inhibitors of CDK2

Type	Name	SMILES
CTDDG	mol1	CC(=O)Nc1cccc(C2=CC3=C4C(=C(C=NN4c4ccc(F)cc4F)NC(=O)c4ccc(nc4)CNc4nc(=O)nc3n4C(C)C)O2)c1
	mol2	NC(=O)c1nc(-c2ccccn2)ccc1Nc1nccc(-c2ccnc3[nH]cc(-c4ccccc4O)c23)n1
	mol3	Cc1cccc2cc(-c3nnc(Nc4ccc(F)c(NC(=O)c5cccn5C)n4)o3)c(=O)[nH]c12
	mol4	Fc1ccc(C2(c3nc4ccccc4[nH]3)NN=C3c4ccc(F)cc4OCC32)cc1
	mol5	CC(=O)Nc1ccc(S(=O)(=O)Oc2cccc3c(Nc4cc(C)c(N)nn4)c4ccccc4nc23)cc1
	mol6	Cc1ccc(N2C(=O)C(N3C(=O)NC(c4ccccc4C(F)(F)F)C3=O)Cc3cc([N+](=O)[O-])ccc32)cc1
	mol7	O=C1NC2CCC(O)CC2c2cc(-n3ccc(Oc4cccc(F)c4)cc3=O)ccc21
	mol8	NC(=O)c1ccnc(Nc2ccc3nc(N4CCCCCC4)n(C(=O)C4=CCCC4)c3c2)c1
	mol9	O=C(Cc1ncc(C(F)(F)F)cc1-c1cccc(OC(F)(F)F)c1)Nc1ccc(S(=O)(=O)N2CCOCC2)cc1
	mol10	NC(=O)c1cc(CCc2c(-c3ccc4cnccc4c3)c(N)n3ccncc23)ccc1O
known inhibitor	Staurosporine	CN[C@@H]1C[C@H]2O[C@@](C)([C@@H]1OC)n1c3ccccc3c3c4c(c5c6ccccc6n2c5c31)C(=O)NC4
	Dinaciclib	CCc1cnn2c(NCc3ccc[n+]([O-])c3)cc(N3CCCC[C@H]3CCO)nc12
	Roscovitine	CC[C@H](CO)Nc1nc(NCc2ccccc2)c2ncn(C(C)C)c2n1

Name of molecules	**Hydrogen Bonds**	**Hydrophobic Interactions**
mol3	**LEU83**, GLN85	ILE10, ALA18, ALA31, LYS33, PHE80
mol7	GLU12	VAL18
Roscovitine	ASP145, **LEU83**	ILE10, VAL18, ALA31, PHE80, PHE82

Table S2: Hydrogen bonds and hydrophobic interactions between inhibitors (mol3, mol7 and Rpscovitine) and CDK2 in the final frame of the 50 ns MD simulation.

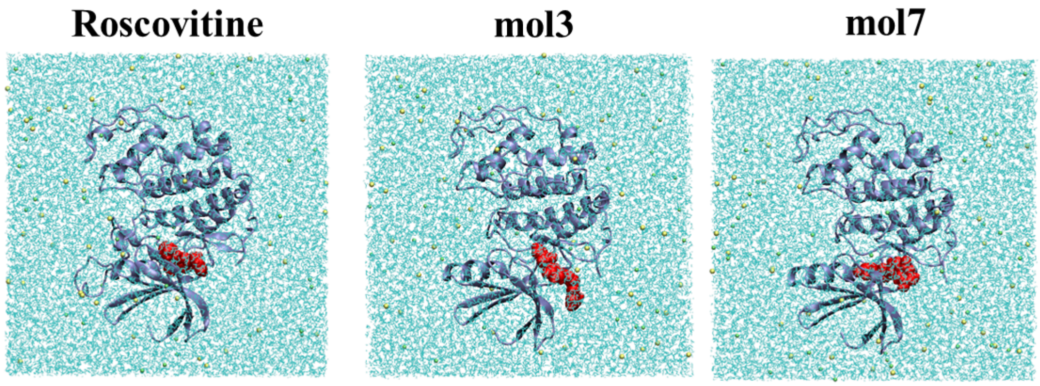

Figure S1: Initial conformations of the molecular dynamics simulations for (left-panel) CDK2-rosocovitine complex (PDB-ID 8FP0), (middle-panel) CDK2-mol3 complex and (right-panel) CDK2-mol7 complex. The complex structures of CDK2-mol3 and CDK2-mol7 were obtained through molecular docking. The CDK2 protein is shown in grey, with inhibitors in red. Water molecules, Na+, and Cl- are depicted in cyan, yellow, and green, respectively.

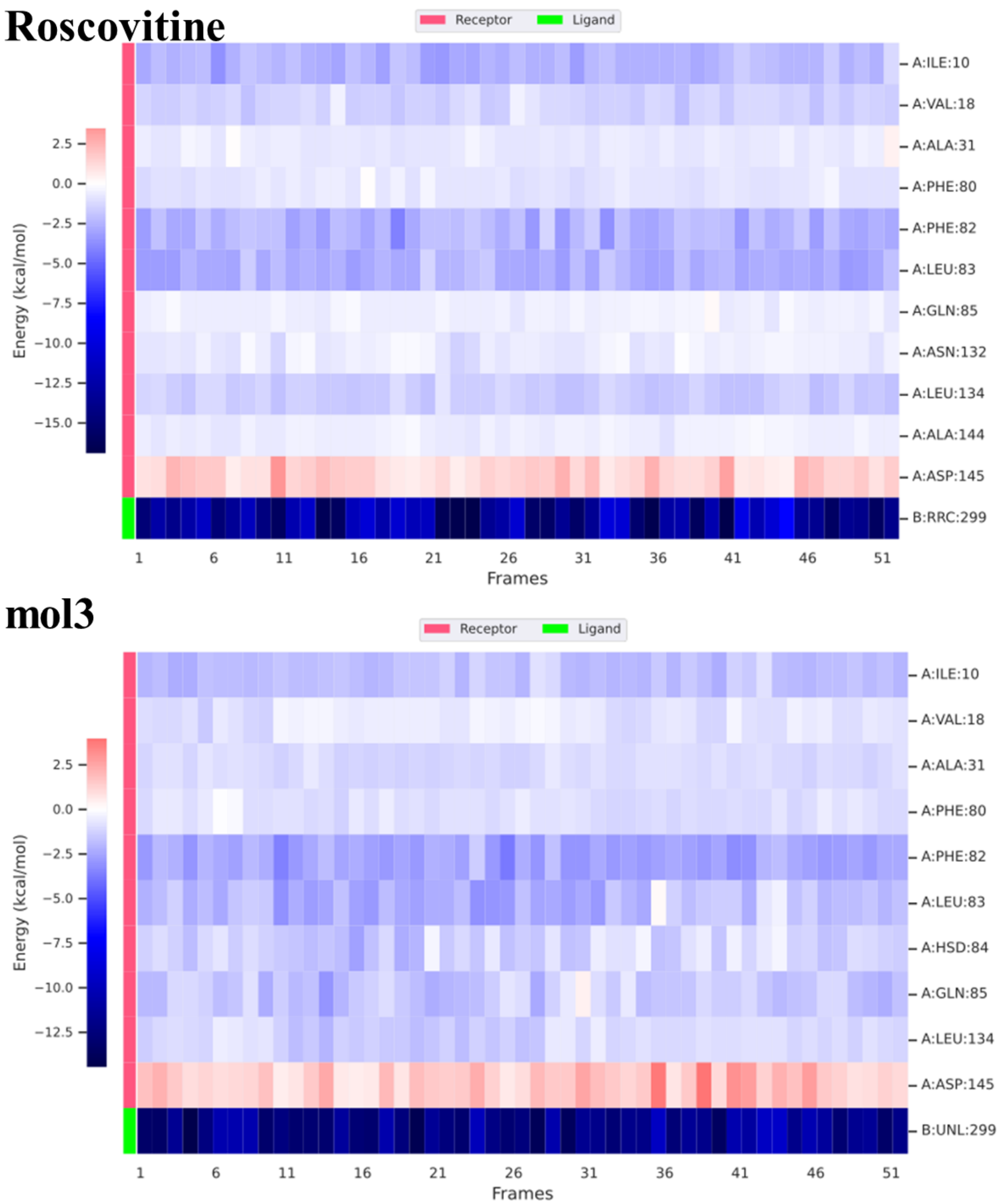

Figure S2: Per-residue binding energy decomposition analysis of Roscovitine (upper panel) and mol3 (lower panel) using the MM-PBSA approach from the last 5 ns of the Molecular Dynamics Simulation trajectory.

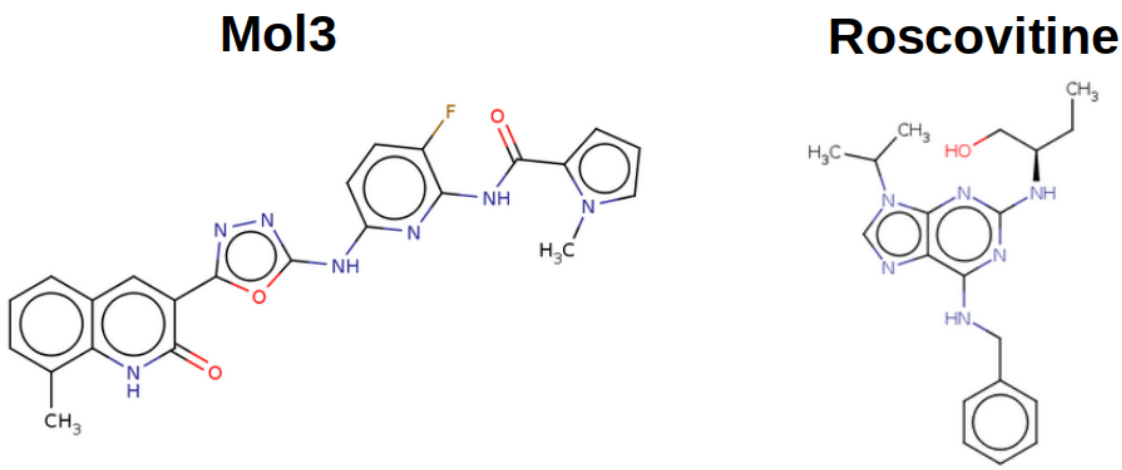

Figure S3: 2D structure of mol3 and Rosocovitine